Nitin B S

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Final Year Computer Science Engineering student From VVCE, Mysuru, with interest in backend engineering — distributed pipelines, APIs, and production ready deployments.

Education

- Vidyavardhaka College Of engineering, Mysuru (B.E. in CS) | 2023-Present — CGPA: 9.09
- Sri Jayachamarajendra PU College, Mysuru (P.U.) — Percentage: 89.5%
- Acharya Vidya kula (Secondary Education, ICSE) — Percentage: 89.1%

Skills

- **Languages:** C, C++, Python
- **Developer Tools:** Git, VS Code, Claude Code, Antigravity, Docker, Kubernetes.
- **Technical Skills:** Data structures, Algorithms, Computer Networks, Operating System, Software Engineering, DBMS

Experience

HPE Career Preview Program (CPP) | Search Engine for Object Storage System — Feb 2026 – Jul 2026

- Built a production-grade RAG pipeline over MinIO object storage, combining dense vector (kNN) and BM25 keyword retrieval via OpenSearch as the vector database, with Redis caching and Kafka-driven event ingestion — deployed on Docker and Kubernetes.
- Engineered an asynchronous document processing pipeline that captures MinIO upload events via Kafka, extracts text/metadata using Apache Tika, and utilizes NLP models (SentenceTransformers/LangChain) to chunk and embed data before indexing it into OpenSearch.

Projects

3D Warehouse Visualization and Vacancy Optimization System

- Built a full-stack slotting engine (FastAPI + async Postgres + Redis/arq) that turns Odoo-shaped inventory data into optimized rack-slot assignments, with a background worker decoupling long-running jobs from the API.
- Implemented pluggable heuristic algorithms across two goals: 3D bin-packing (FFDH+CoM, BLF+weight stratification) feeding a Storage Location Assignment stage (W-BFDH), plus 3 independent picking-efficiency strategies (golden-zone, affinity clustering, S-shape routing).
- Centralized safety-constraint validation (clearance, weight capacity, center-of-mass, aisle density) as shared checks every algorithm must pass before a placement is accepted.
- Built a React Three Fiber 3D digital twin for live rack occupancy, with an exception queue and manual-override workflow, backed by versioned config for reproducible runs.

AI-Powered YouTube Shorts Automation System

- Engineered a fully autonomous Python monorepo that orchestrates scriptwriting, voiceovers (Edge-TTS), and video assembly (MoviePy/FFmpeg) to publish content across a YouTube channel.
- Architected a serverless infrastructure by leveraging GitHub Actions for scheduling, a Streamlit dashboard for monitoring the project, and multiple APIs.
- Designed a dual-track LLM-in-the-loop validation system with a prompt-engineered safety gate that auto-screens AI-generated content for appropriateness before publish — demonstrating multi-step agentic coordination and responsible AI deployment.

Extracurricular Activities

- Volunteer Teacher | U&I (NGO)
- Peer tutor
- Coordinator | Programming Workshop by Department